

1 2 NOV 2025

B.C.A Semester-3 Examination

Seat No: _____

DATA AND FILE STRUCTURE – I [306]

Subject Code: 25739-[BCAM206-2C]

Time: 01:15 Hours

Total Marks: 35

Q-1 What is Sparse Matrix? Explain Representation of Sparse Matrix with example. [10]

OR

Q-1 [A] Explain Binary Search with example. [05]

Q-1 [B] Write an algorithm/program of insertion sort. [05]

Q-2 What is Double Linked List? Explain any two operations with example. [10]

OR

Q-2 [A] Write a short note on Circular Linked List. [05]

Q-2 [B] Difference between Single Linked List and Double Linked List. [05]

Q-3 What is Queue? Explain its all operations. [10]

OR

Q-3 [A] Write a short note on Stack. [05]

Q-3 [B] Define the term “infix”, “prefix”, “postfix”. [05]

Q-4 Do as Directed. [Any 5 out of 6] [05]

1. what is linear search?
2. Define the term “Data Structure”.
3. A Stack is called a FIFO Data Structure. [True / False]
4. List out different application of Queue.
5. The address of the first node in a Linked List is stored in a separate location/pointer is called the head. [True / False]
6. Binary search is always faster than linear search. [True / False]